

CHI WAI CHAN

Hong Kong

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Summary

Software engineer with 3+ years of experience building cloud automation and observability solutions across Juju, OpenStack, and Kubernetes ecosystems. Open-source contributor with hands-on expertise in developing charmed operators, exporters, and tooling to integrate with Grafana and Prometheus. Strong engineering foundation in Go and Python, with working knowledge of JavaScript and modern CI/CD practices.

Experience

Canonical

May 2022 – Present

Graduate DevOps Engineer, Associate DevOps Engineer, Software Engineer I

Hong Kong, China

- Progressed from Graduate DevOps Engineer to Software Engineer I, taking on broader ownership across cloud operations tooling and platform features.
- Implemented Falco and Falcosidekick operators to enable cloud operators to deploy and monitor runtime security across OpenStack and Kubernetes environments.
- Built Sunbeam validation using Tempest and integrated periodic validation results into Grafana with alerts to help operators detect regressions earlier.
- Delivered maintenance workflows for Sunbeam and MicroCeph to reduce operational burden when maintaining OpenStack and Ceph nodes.
- Built and shipped Sunbeam observability with Grafana dashboards and alerts to improve day-2 operations for OpenStack deployments.
- Developed prometheus-hardware-exporter (Python-based) to provide vendor-specific RAID metrics and improve bare-metal visibility.
- Delivered the Hardware Observer operator with COS Lite integration, providing a modern monitoring stack for Juju-managed bare-metal infrastructure.
- Unblocked QA testing pipelines by fixing critical issues in Graylog and Elasticsearch charms.
- Improved charm reliability by adding unit/functional tests and implementing clean-up in apt-mirror to help operators reduce unused mirrored storage.
- Built prometheus-juju-backup-all-exporter (Snap-based) and integrated it with COS Lite to make backup operations observable.

Research Center for the Early Universe, University of Tokyo

Aug 2021 – Apr 2022

Project Researcher

Tokyo, Japan

- Improved GstLAL pipeline's offline dashboard to support on-demand loading of images.
- Developed a static group website and prepared TensorFlow tutorial for new members in the group.
- Helped re-analyze gravitational wave with the latest GstLAL pipeline at the time, and ensured the quality of the pipeline.

HENNGE

Jul 2021 – Aug 2021

Software Engineer Intern

Tokyo, Japan

- Contributed a social media web application using Python (Flask framework).
- Implemented authentication and login system with Google OAuth 2.0 and Redis.
- Implemented text and image storage system using Redis and MinIO.
- Employed micro-services architecture and deployed the web application using Docker, AWS, and Terraform.

Education

The University of Tokyo <i>Master of Science in Physics</i>	Sep 2018 – Sep 2020 <i>Tokyo, Japan</i>
• Overall grade A in course work performance • Teaching Assistant in Gravitational Wave Physics • Submitted a technical paper to arXiv (arXiv:2009.03025)	
The Hong Kong University of Science and Technology <i>Bachelor of Science in Physics</i>	Sep 2014 – Sep 2018 <i>Hong Kong</i>
• Dean's List 2015 - 2017 • HKUST Robust team (Smart Car team 2016 - 2017) • Graduated with First Honor Class (GPA 3.7/4.3)	

Technical Skills

• Falco • Docker • Kubernetes • OpenStack	• Terraform • Juju • ROCK • LXD	• Snap • Grafana • Prometheus • Loki	• Git • Linux • CI/CD • Scrum	• Python • Go • JavaScript
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Certifications

AWS Certified Cloud Practitioner Amazon Web Services Training and Certification	December 2025 - December 2028 LF-ehjghdyant
Certified Kubernetes Administrator The Linux Foundation	October 2025 - October 2027 LF-effnrqhxpf
Linux Foundation Certified Systems Administrator The Linux Foundation	June 2025 - June 2027 LF-ehjghdyant
Certified Kubernetes Application Developer The Linux Foundation	Nov 2024 - Nov 2026 LF-8023d8bwns

Achievements

Masanori Nagashima Scholarship	2019
Dr Sampson Sun Scholarship Fund	2018
Third place winner at 12th NXP Cup Intelligent Car Racing Competition (South China Region)	2017

Languages

• Cantonese	• English	• Mandarin	• Japanese (N2 level)
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